



Question # 1 of 10 (Start time: 04:53:00 PM, 14 December 2022)

The family of curve

$$x^2 + y^2 = C^2$$

represents a family of circles centered at _____.

Select the correct option

- | | |
|-----------------------|-------|
| ☐ | (1,1) |
| ☐ | (0,0) |
| ☐ | (1,0) |
| ☐ | (0,1) |

Click to :



MTH401 - Differential Equations (Quiz#1)

Question # 1 of 10 (Start time: 05:04:46 PM, 14 December 2022)

In first order chemical reaction,

Select the correct option

☐

the molecules of the substance remain same.

☐

None of them

☐

the molecule of the substance makes smaller molecules.

☐

the molecule of the substance makes large molecules.

Click to Save Ans



Question # 3 of 10 (Start time: 04:56:34 PM, 14 December 2022)

Let

$$f_1(x) = e^{m_1 x}$$

and

$$f_2(x) = e^{m_2 x}$$

are linearly independent, because

$$c_1 f_1(x) + c_2 f_2(x) = 0$$

iff

Select the correct option



☐	$c_1 = c_2 = 0, m_1 \neq m_2$
☐	$c_1 = c_2 = 0, m_1 = m_2$
☐	$c_1 = c_2 \neq 0, m_1 = m_2$
☐	$c_1 = c_2 \neq 0, m_1 \neq m_2$

Click to Save Answer &



Q10 (Start time: 04:54:40 PM, 14 December 2022)

If T be the half - life then solution of the population growth model could be _____

Correct option



$$\frac{A_0}{2}$$

$$\frac{A_0}{2} = A_0 e^{kt}$$

$$\frac{A_0}{2} (A_0 e^{kt})$$

$$A_0 e^{kt}$$

Click to Save Answer &



Question # 4 of 10 (Start time: 04:59:00 PM, 14 December 2022)

In the logistic equation

$$\frac{1}{p} \frac{dp}{dt} = a - bp$$

the term

$$-bp^2$$

can be interpreted as

Select the correct option

☐	inhibition term
☐	unconstraint term
☐	constraint term
☐	self-abandonment term

Click to Save Answer

1 of 10 (Start Date: 05/05/20 PM, 14 December 2022)

$$f_1(x) = e^{m_1 x}$$

$$f_2(x) = e^{m_2 x}$$

only independent functions

$$c_1 f_1(x) + c_2 f_2(x) = 0$$

is correct option

Related Math Equ

$$c_1 = c_2 = 0, m_1 \neq m_2$$

$$c_1 = c_2 = 0, m_1 = m_2$$

Select the correct option

Reload M

- ☐ $c_1 = c_2 = 0, m_1 \neq m_2$
- ☐ $c_1 = c_2 = 0, m_1 = m_2$
- ☐ $c_1 = c_2 \neq 0, m_1 \neq m_2$
- ☐ $c_1 = c_2 \neq 0, m_1 = m_2$

Click to Save Answer & Move to Next Question

EN 12/14

Start time: 05:07:10 PM, 14 December 2022)

For non-homogeneous differential equation

$$a_2(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

Following is NOT the pair of boundary condition _____

option



Reload Math E

$$y'(a) = y'_0, \quad y'(b) = y'_0$$

$$y'(a) = y'_0, \quad y(b) = y_1$$

$$y'(a) = y'_0, \quad y'(b) = y'_1$$

$$y'(a) = y'_0, \quad y(b) = y_1$$

Click to Save Answer & Move to Next Question

EN

5:07
12/14



Question # 9 of 10 (Start time: 05:05:48 PM, 14 December 2022)

if

$$a_n(x) = 0$$

in the differential equation

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

for some

$$x \in I$$

then solution of initial value problem may NOT be_____.

Select the correct option

☐	exist
☐	none of these
☐	unique
☐	exist and unique

Click to Save Answer

Question # 3 of 10 (Start time: 05:07:10 PM, 14 December 2022)

For 2nd order linear non-homogeneous differential equation

$$a_2(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

, which of the following is NOT the pair of boundary condition _____

Select the correct option

Reload

- | | |
|-----------------------|------------------------------------|
| ☐ | $y'(a) = y'_0, \quad y'(b) = y'_0$ |
| ☐ | $y'(a) = y'_0, \quad y(b) = y_1$ |
| ☐ | $y'(a) = y'_0, \quad y'(b) = y'_1$ |
| ☐ | $y'(a) = y'_0, \quad y(b) = y_1$ |

Click to Save Answer & Move to Next Q

EN

Quiz

x +

vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=3921e3cf-8632-47f2-a864-f4ad25a04f7f



H401 - Differential Equations (Quiz#1)

Quiz Start Time: 05:04 PM, 14 December 2022

Question # 4 of 10 (Start time: 05:08:11 PM, 14 December 2022)

Total Marks

If T be the half-life then solution of the population growth model could be _____.

Select the correct option

Reload Math Equations

- ☐ $\frac{A_0}{2} (A_0 e^{kt})$
- ☐ $\frac{A_0}{2}$
- ☐ $A_0 e^{kt}$
- ☐ $\frac{A_0}{2} = A_0 e^{kt}$



New Tab - Google



Quiz - Google

Click to Save Answer & Move to Next Question

EN

5:08 PM

12/14/2022



MTH401 – Differential Equations (Quiz#1)

Qu

Question # 10 of 10 (Start time: 05:08:26 PM, 14 December 2022)

If Wronskian $W(h, z)(x) = 0$ then

Select the correct option

☐ $h(x)$ and $z(x)$ are linearly independent.☐ $h(x)$ and $z(x)$ are linearly dependent.

Click t

Question # 7 of 10 (Start time: 05:11:18 PM, 14 December 2022)

The family of curve

$$x^2 + y^2 = C^2$$

represents a family of circles centered at _____.

Select the correct option

- | | |
|-----------------------|-------|
| ☐ | (1,1) |
| ☐ | (0,1) |
| ☐ | (1,0) |
| ☐ | (0,0) |



New Tab - Googl...



Quiz - Google Chr...

Click



MTH401 - Differential Equations (Quiz#1)

Quiz S

Question # 6 of 10 (Start time: 05:10:34 PM, 14 December 2022)

In cooling/heating differential equation $\frac{dT}{dt} = k(T - T_m)$, the term T_m represents _____

Select the correct option

- | | |
|-----------------------|--|
| ☐ | Change in temperature |
| ☐ | Specific constant of heat |
| ☐ | Temperature of surrounding environment |
| ☐ | Temperature of body |



New Tab - Googl...



Quiz - Google Chr...

Click to Save Answer

Question # 5 of 10 (Start time: 05:09:18 PM, 14 December 2022)

The differential equation of orthogonal trajectory to the family of curves

$$x^2 + y^2 = C^2$$

is

Select the correct option

- | | |
|-----------------------|--------------------------------|
| ☐ | $\frac{dy}{dx} = -\frac{x}{y}$ |
| ☐ | $\frac{dy}{dx} = -\frac{y}{x}$ |
| ☐ | $\frac{dy}{dx} = \frac{x}{y}$ |
| ☐ | $\frac{dy}{dx} = \frac{y}{x}$ |



New Tab - Google...



Quiz - Google Chr...

Click to Save Answer & M...

B

Question # 8 of 10 (Start time: 05:12:17 PM, 14 December 2022)

The general solution of non-homogenous equation is,

Select the correct option

☐	$y_p + y_c$
☐	$y_c + y_p$
☐	$y_p - y_c$
☐	$y_c - y_p$



New Tab - Googl...



Quiz - Google Chr...

Question # 9 of 10 (Start time: 05:13:32 PM, 14 December 2022)

If initial amount of a radioactive isotope is 100g. What will be the amount at the end of 30 days such that $K=0.043$?

Select the correct option

☐

371.415

☐

383.279

☐

360.351

☐

380.560

Click to Save Answer & Move to Next Q

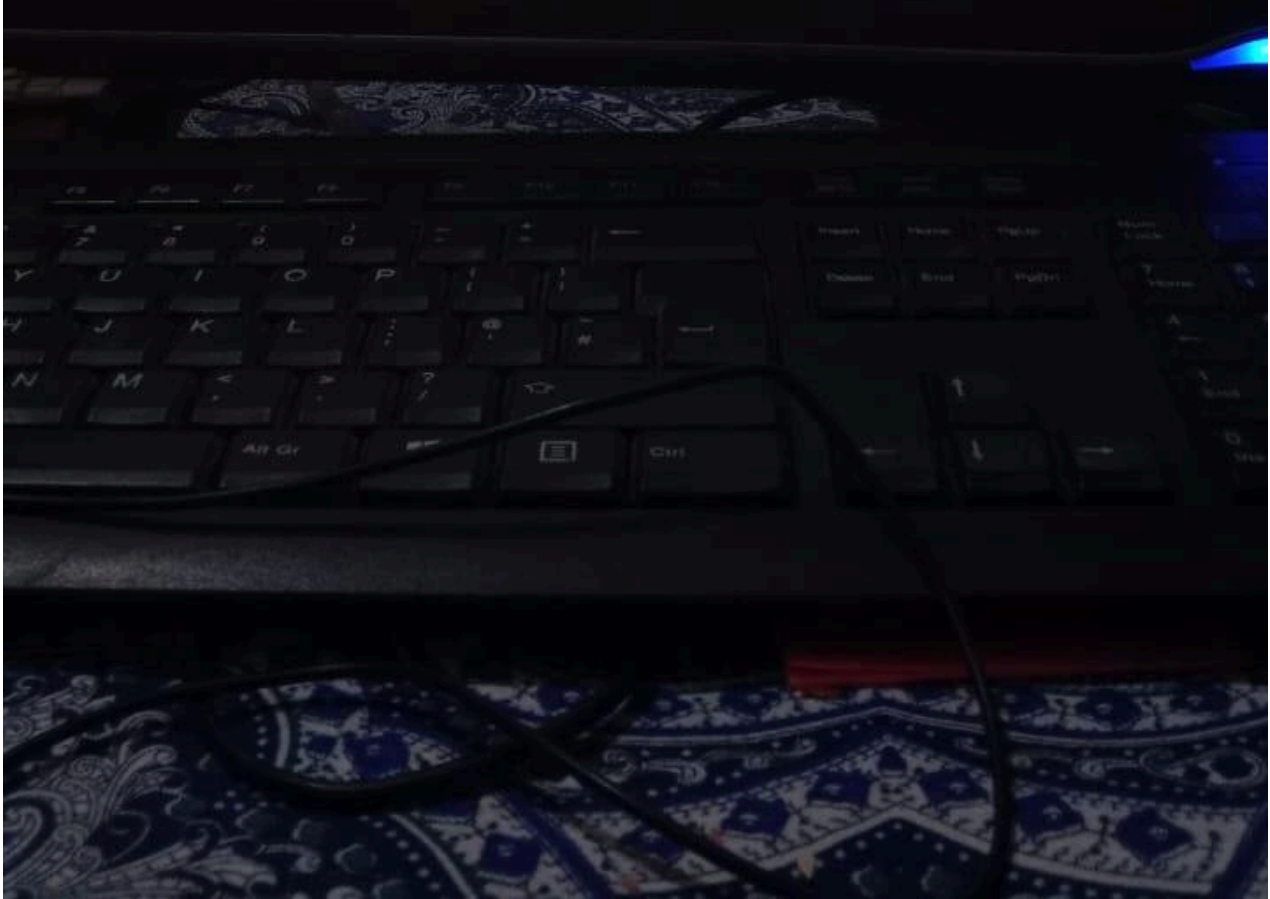
EN

A function satisfying the differential equation on the some interval I , containing a and b , whose graph passes through two points (a, y_0) and (b, y_1) is called solution of the _____ value problem.

the correct option

Reload Math Equations

☐	boundary
☐	initial



Question # 2 of 10 (Start time: 05:15:42 PM, 14 December 2022)

Total Marks: 1

A set of functions

$$\{f_1(x), f_2(x), f_3(x), \dots, f_n(x)\}$$

is said to be _____ on an interval I if there exist constants

$$c_1, c_2, c_3, \dots, c_n$$

not all zero, such that

$$c_1 f_1(x) + c_2 f_2(x) + \dots + c_n f_n(x) = 0, \quad \forall x \in I$$

Select the correct option

 Reload Math Equations

linear dependent

☐

linear independent

☐

Click to Save Answer & Move to Next Question

sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 05:14 PM, 14 December 2022

Question # 1 of 10 (Start time: 05:14:50 PM, 14 December 2022)

Total Marks: 1

Orthogonal trajectories occur naturally in the area(s) of _____.

Select the correct option

☐

(i) physics

☐

(iv) All (i), (ii) and (iii).

☐

(iii) electricity and magnetism

☐

(ii) fluid dynamics

Click to Save Answer & Move to Next Question

Question # 4 of 10 (Start time: 05:17:48 PM, 14 December 2022)

Total Marks: 1

Let

$$f_1(x) = e^{m_1 x}$$

and

$$f_2(x) = e^{m_2 x}$$

are linearly independent because

$$c_1 f_1(x) + c_2 f_2(x) = 0$$

iff

Select the correct option

 Reload Math Equations


$$c_1 = c_2 = 0, m_1 \neq m_2$$



$$c_1 = c_2 \neq 0, m_1 = m_2$$



$$c_1 = c_2 \neq 0, m_1 \neq m_2$$



$$c_1 = c_2 = 0, m_1 = m_2$$

Click to Save Answer & Move to Next Question

Question # 3 of 10 (start time: 05:16:59 PM, 14 December 2022)

Total Marks: 1

A set of n functions

$$\{f_1(x), f_2(x), f_3(x), \dots, f_n(x)\}$$

is _____ if at least one of them can be expressed as a linear combination of the remaining.

Select the correct option

 Reload Math Equations

linear independent

☐

linear dependent

☐

Click to Save Answer & Move to Next Question



MTH401 - Differential Equations (Quiz#1)

Total Questions :10

Please read the following instructions carefully!

1. Quiz will be based upon Multiple Choice Questions (MCQs).
2. You have to attempt the quiz online. You can start attempting the quiz any time within given date(s) of a particular subject by clicking the link for Quiz in VULMS.
3. Each question has a fixed time of 90 seconds. So you have to save your answer before 90 seconds. But due to unstable internet speeds, it is recommended that you should save your answer within 50 seconds. While attempting a question, keep an eye on the remaining time.
4. Attempting quiz is unidirectional. Once you move forward to the next question, you can not go back to the previous one. Therefore before moving to the next question, make sure that you have selected the best option.
5. **DO NOT** press **Back Button / Backspace Button** while attempting a question, otherwise you will lose that question.
6. **DO NOT** refresh the page unnecessarily, **specially** when following messages appear
Saving...
Question Timeout: Now loading next question...
7. **Javascript MUST be enabled** in your browser; otherwise you will not be able to attempt the quiz.
8. If for any reason, you lose access to internet (like power failure or disconnection of internet), you will be able to attempt the quiz again from the question next to the last shown question. But remember that you have to complete the quiz before expiry of the deadline.
9. If any student failed to attempt the quiz in given time then no re-take or offline quiz will be held.

Important Announcement

1. It has come to the university's knowledge that some students are using unfair means while attempting their quiz.
2. The University is monitoring & analyzing the quiz attempts and if a student is found involved in using unfair means, i.e., helping tools/software, Web Extension, Add-ons etc., an Unfair Means Case (UMC) will be made against such student(s) including cancellation of their quiz.

Start Quiz

The Wronskian of two functions

$f(x)$

and

$g(x)$

is:

Select the correct option

 Reload Math Equations

☐	$f(x)g'(x) - f'(x)g(x)$
☐	$f'(x)g'(x) - f(x)g(x)$
☐	$f'(x)g'(x) + f(x)g(x)$
☐	$f(x)g'(x) + f'(x)g(x)$

Click to Save Answer & Move to Next Question

Question # 1 of 10 (Start time: 07:00:08 PM, 14 December 2022)

Total Marks: 10

A set of functions

$$\{f_1(x), f_2(x), f_3(x), \dots, f_n(x)\}$$

is said to be _____ on an interval I if there exist constants

$$c_1, c_2, c_3, \dots, c_n$$

not all zero, such that

$$c_1 f_1(x) + c_2 f_2(x) + \dots + c_n f_n(x) = 0, \quad \forall x \in I$$

Select the correct option



Reload Math Equations

linear independent

☐

linear dependent

☐

Click to Save Answer & Move to Next Question

Question # 8 of 10 (Start time: 07:08:35 PM, 14 December 2022)

Total Marks: 1

A set of functions

$$\{f_1(x), f_2(x), f_3(x), \dots, f_n(x)\}$$

is said to be _____ on an interval I if there exist constants

$$c_1, c_2, c_3, \dots, c_n$$

are all zero, such that

$$c_1 f_1(x) + c_2 f_2(x) + \dots + c_n f_n(x) = 0, \quad \forall x \in I$$

Select the correct option



Reload Math Equations

linear independent

☐

linear dependent

☐

Click to Save Answer & Move to Next Question

Orthogonal trajectories occur naturally in the area(s) of_____.

Select the correct option

☐

(ii) fluid dynamics

☐

(i) physics

☐

(iv) All (i), (ii) and (iii).

☐

(iii) electricity and magnetism

Click to Save Answer & Move to Next Question

Question # 9 of 10 (Start time: 07:10:00 PM, 14 December 2022)

Total Marks: 1

Wronskian

$$W(1, e^x)$$

= _____

Select the correct option

 Reload Math Equations

☐

1

☐

$$e^x$$

☐

-1

☐

0

Click to Save Answer & Move to Next Question

If $A(t)$ is the amount of a radioactive substance present at time t , the the rate of change of $A(t)$ with respect to t is given by _____.

Select the correct option

$$A_0 e^t$$

$$dA = dt$$

$$\frac{A_0}{2}$$

$$\frac{dA}{dt} = kA; k \text{ is a constant of proportionality}$$

Click to Show Answer & Hint

if

$$a_n(x) = 0$$

in the differential equation

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

for some

$$x \in I$$

then solution of initial value problem may NOT be_____.

Select the correct option

 Reload Math Equations

☐

none of these

☐

exist and unique

☐

unique

☐

exist

Click to Save Answer & Move to Next Question

Question # 10 of 10 (start time: 07:10:33 PM, 14 December 2022)

Total Marks: 1

In first order chemical reaction ,

Select the correct option

 Reload Math Equations☐

None of them

☐

the molecule of the substance makes smaller molecules.

☐

the molecule of the substance makes large molecules.

☐

the molecules of the substance remain same.

Click to Save Answer & Move to Next Question



The two functions $u(y)$ and $v(y)$ are linearly independent if Wronskian $W(u,v)$ is...

Select the correct option



not equal to zero



equal to zero



greater than zero



less than zero

Click to Save Answer



BC190406254: TAMOOR JAMIL

Time Left

89
sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 11:12 PM, 14 December 2022

Question # 5 of 10 (Start time: 11:15:55 PM, 14 December 2022)

Total Marks: 1

If initial amount of a radioactive isotope is 100g. What will be the amount at the end of 30 days such that $K=0.043$?

Select the correct option

- | | |
|-----------------------|---------|
| ☐ | 360.351 |
| ☐ | 363.279 |
| ☐ | 380.560 |
| ☐ | 371.415 |

Click to Save Answer & Move to Next Question

The leading coefficient in the differential equation:

$$3x \frac{d^2 y}{dx^2} - 7x \frac{dy}{dx} + 5y = 0$$

Select the correct option

 Reload Math Equations

☐

-7

☐

3

☐

7

☐

5

Click to Save Answer & Move to Next Question

MTH401 – Differential Equations (Quiz#1)

Question # 1 of 10 (Start time: 08:37:57 PM, 14 December 2022)

The homogeneous linear differential equations always possess the _____ solution.

Select the correct option

☐	trivial
☐	non-trivial



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36



BC190406254: TAMOOR JAMIL

Time Left

87
sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 11:12 PM, 14 December 2022

Question # 9 of 10 (Start time: 11:21:56 PM, 14 December 2022)

Total Marks: 1

Orthogonal trajectories occur naturally in the area(s) of _____.

Select the correct option



(iv) All (i), (ii) and (iii).



(ii) fluid dynamics



(iii) electricity and magnetism



(i) physics

Click to Save Answer & Move to Next Question

Question # 3 of 10 (Start time: 08:39:23 PM, 14 December 2022)

A function satisfying the differential equation on the some interval I , containing a and b , whose graph passes through two points (a, y_0) and (b, y_1) is called solution of the _____ value problem.

Select the correct option

Reload Math Eq

☐ initial☐ boundary

Clicking Stop Answer & move to Next Question

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 08:37 PM, 14

Question # 2 of 10 (Start time: 08:38:34 PM, 14 December 2022)

A set of n functions

$$\{f_1(x), f_2(x), f_3(x), \dots, f_n(x)\}$$

is _____ if at least one of them can be expressed as a linear combination of the remaining.

Select the correct option

☐	linear independent
☐	linear dependent



BC200404637: FATIMA ASHRAF

Time Left 89
sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 08:35 PM, 14 December 2022

Question # 5 of 10 (Start time: 08:40:16 PM, 14 December 2022)

Total Marks: |

If X is the remaining amount of the substance A at any time t then, $\frac{dX}{dt} = kX$, where $k < 0$ because X is _____.

Select the correct option

Reload Math Equations

☐	increasing
☐	decreasing

Click to Save Answer & Move to Next Question

Question # 5 of 10 (Start time: 05:19:07 PM, 14 December 2022)

Total Marks: 1

If we substitute

$$u = \frac{y}{x}$$

in differential equation

$$x \frac{dy}{dx} - y = \frac{x^3}{y} e^{\frac{y}{x}}$$

then the given differential equation can be simplified to

Select the correct option

 Reload Math Equations

☐

$$ue^u du = dy$$

☐

$$ue^u du = dx$$

☐

$$ue^{(-u)} du = dx$$

☐

$$ue^{(-u)} du = dy$$

Click to Save Answer & Move to Next Question



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3



BC210403078: ABDUL RASHEED

Time Left 89 sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 10:59 PM, 14 December 2022

Question # 6 of 10 (Start time: 11:06:00 PM, 14 December 2022)

Total Marks: 1

In exponential growth for population growth

$$N(t) = N_0 e^{mt}$$

If $m < 0$ then

$$\lim_{t \rightarrow \infty} N(t) =$$

Select the correct option

Reload Math Equations

☐	$-\infty$
☐	∞
☐	0
☐	1

Click to Save Answer & Move to Next Question



Question # 7 of 10 (Start time: 11:02:14 PM, 14 December 2022)

A set of functions

$$\{f_1(x), f_2(x), f_3(x), \dots, f_n(x)\}$$

is said to be _____ on an interval I if there exist constants

$$c_1, c_2, c_3, \dots, c_n$$

are all zero, such that

$$c_1 f_1(x) + c_2 f_2(x) + \dots + c_n f_n(x) = 0, \quad \forall x \in I$$

Select the correct option

☒ linear dependent

☐ linear independent

Click to Save Answer



BC200404637: FATIMA ASHRAF

Time Left 87
sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 08:35 PM, 14 December 2022

Question # 6 of 10 (Start time: 08:41:47 PM, 14 December 2022)

Total Marks: 1

The family of curve

$$x^2 + y^2 = C^2$$

represents a family of circles centered at _____

Select the correct option

Reload Math Equations

☐	(0,0)
☐	(0,1)
☐	(1,1)
☐	(1,0)

Click to Save Answer & Move to Next Question

Question # 4 of 10 (Start time: 10:03:32 PM, 14 December 2022)

A function satisfying the differential equation on the interval I whose graph passes through the point (x_0, y_0) and has the slope at the point (x_0, y_0) is the number y'_0 is called solution of the differential equation.

Select the correct option

boundary

initial

Activate Windows

Go to Settings to activate Windows.



12°C





Quiz

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BC200414970: AMNA

Time Left 88
sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 04:42 PM, 14 December 2022

Question # 3 of 10 (Start time: 04:43:46 PM, 14 December 2022)

Total Marks: 1

The differential equation

$$\frac{d^2 u}{dx^2} - 2xu + \sin x = 0$$

is

Select the correct option

 Reload Math Equations

linear and non-homogenous



linear and homogenous



non-linear and non-homogenous



non-linear and homogenous

Click to Save Answer & Move to Next Question



Question # 8 of 10 (start time: 08:46:01 PM, 14 December 2022)

In exponential model for the population growth

$$P(t) = P_0 e^{kt} \quad \text{If } k < 0, \text{ then } \lim_{t \rightarrow \infty} P(t) =$$

Select the correct option

- | | |
|-----------------------|-----------|
| ☐ | $-\infty$ |
| ☐ | ∞ |
| ☐ | 0 |
| ☐ | 1 |



Question # 6 of 10 (Start time: 08:42:28 PM, 14 December 2022)

A set of functions

is said to be _____ on an interval I if there exist constants

not all zero, such that

$$\{f_1(x), f_2(x), f_3(x), \dots, f_n(x)\}$$

$$c_1, c_2, c_3, \dots, c_n$$

$$c_1 f_1(x) + c_2 f_2(x) + \dots + c_n f_n(x) = 0, \quad \forall x \in I$$

Select the correct option

☐ linear independent

☐ linear dependent

Question # 7 of 10 (Start time: 08:43:40 PM, 14 December 2022)

The differential equation

$$\frac{d^2 u}{dx^2} - 2xu + \sin x = 0$$

is:

Select the correct option

- ☐ non-linear and homogenous
- ☐ linear and homogenous
- ☐ linear and non-homogenous
- ☐ non-linear and non-homogenous



BC210420813: MUHAMMAD SAQLAIN

Time Left 88
sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 10:04 PM, 14 December 2022

Question # 5 of 10 (Start time: 10:08:57 PM, 14 December 2022)

Total Marks: 1

A modification of the nonlinear logistic differential equation

$$\frac{1}{p} \frac{dp}{dt} = a - b \ln p$$

has been used in the study of

Select the correct option

Reload Math Equations

- | | |
|-----------------------|--------------------------------------|
| ☐ | revenue growth of commercial product |
| ☐ | solid tumor |
| ☐ | all of these |
| ☐ | actuarial prediction |

Click to Save Answer & Move to Next Question



BC190406254: TAMOOR JAMIL

Time Left 89 sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 11:12 PM, 14 December 2022

Question # 4 of 10 (Start time: 11:15:09 PM, 14 December 2022)

Total Marks: 1

The family of curve

$$x^2 + y^2 = C^2$$

represents a family of circles centered at _____

Select the correct option

Reload Math Equations

- | | |
|-----------------------|-------|
| ☐ | (0,0) |
| ☐ | (1,0) |
| ☐ | (1,1) |
| ☐ | (0,1) |

Click to Save Answer & Move to Next Question

Question # 7 of 10 (Start time: 10:07:09 PM, 14 December 2022)

The two functions $u(y)$ and $v(y)$ are linearly independent if Wronskian $W(u,v)$ is...

Select the correct option

 Reload

☐

not equal to zero

☐

greater than zero

☐

less than zero

☐

equal to zero

Activate Windows
Go to Settings to activate Windows.

The differential equation

$$\frac{d^3y}{dx^3} + y^2 = x^4$$

is _____.

Select the correct option

Reload Math Equations

☐

None of these

☐

Non-homogeneous and linear

☐

Homogeneous and non linear

☐

Non homogeneous and non-linear

Click to Save Answer & Move to Next Question



BC200404637: FATIMA ASHRAF

Time Left 89
sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 08:35 PM, 14 December 2022

Question # 8 of 10 (Start time: 08:44:26 PM, 14 December 2022)

Total Marks: |

The logistic equation, $\frac{dP}{dt} = P(a - bP)$, can be identified as a nonlinear equation that is _____.

Select the correct option

Reload Math Equations

☐	integrable
☐	separable

Click to Save Answer & Move to Next Question



BC200413286: MUHAMMAD ABU BAKAR

MTH401 – Differential Equations (Quiz#1)

Question # 8 of 10 (Start time: 09:01:00 PM, 14 December 2022)

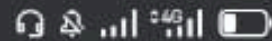
The general solution of non-homogenous equation is,

Select the correct option

☐	$y_p - y_c$
☒	$y_p + y_c$
☐	$y_c - y_p$
☐	$y_c * y_p$

C

10:09



Quiz

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Question # 3 of 10 (Start time: 10:08:24 PM, 14 December 2022)

The Wronskian of two functions

 $f(x)$

and

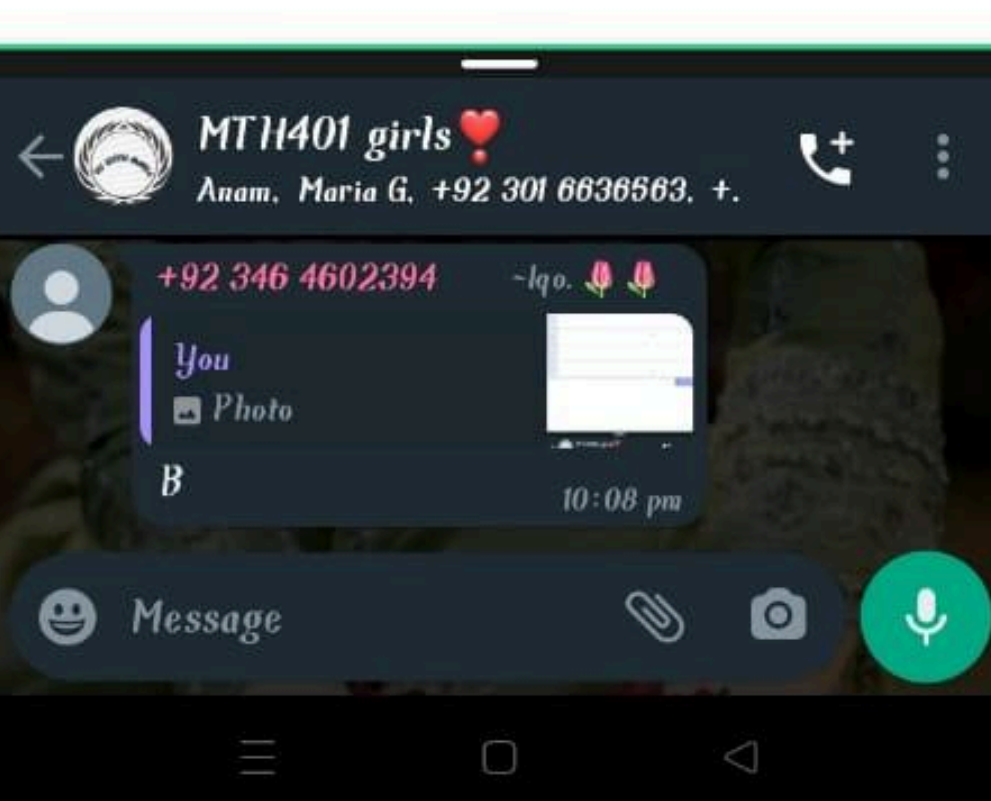
 $g(x)$

is

Select the correct option

- | | |
|-----------------------|-------------------------|
| ☐ | $f'(x)g'(x) + f(x)g(x)$ |
| ☐ | $f'(x)g'(x) - f(x)g(x)$ |
| ☐ | $f(x)g'(x) + f'(x)g(x)$ |
| ☐ | $f(x)g'(x) - f'(x)g(x)$ |

Click



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21



BC210420813: MUHAMMAD SAQLAIN

Time Left 89
sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 10:04 PM, 14 December 2022

Question # 3 of 10 (Start time: 10:06:51 PM, 14 December 2022)

Total Marks: 1

In first order chemical reaction ,

Select the correct option

Reload Math Equations

- | | |
|-----------------------|--|
| ☐ | the molecules of the substance remain same. |
| ☐ | None of them |
| ☐ | the molecule of the substance makes smaller molecules. |
| ☐ | the molecule of the substance makes large molecules. |

Click to Save Answer & Move to Next Question

BC200400580: USMAN SALEEM

Time Left 89 sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 10:49 PM, 14 December 2022

Question # 2 of 10 (Start time: 10:50:18 PM, 14 December 2022)

Total Marks: 1

If Wronskian $W(h,z)(x)=0$ then

Select the correct option

Reload Math Equations

☐ $h(x)$ and $z(x)$ are linearly dependent.

☐ $h(x)$ and $z(x)$ are linearly independent.

Click to Save Answer & Move to Next Question



MTH401 - Differential Equations (Quiz#1)

Question # 7 of 10 (Start time: 08:58:54 PM, 14 December 2022)

Orthogonal trajectories occur naturally in the area(s) of_____.

Select the correct option

- | | |
|-----------------------|---------------------------------|
| ☐ | (ii) fluid dynamics |
| ☐ | (iv) All (i), (ii) and (iii). |
| ☐ | (i) physics |
| ☐ | (iii) electricity and magnetism |



BC190406254: TAMOOR JAMIL

Time Left

89
sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 11:12 PM, 14 December 2022

Question # 3 of 10 (Start time: 11:14:16 PM, 14 December 2022)

Total Marks: 1

In exponential model for the population growth

$$P(t) = P_0 e^{kt} \text{ If } k > 0, \text{ then } \lim_{t \rightarrow \infty} P(t) =$$

Select the correct option

Reload Math Equations

☐	1
☐	$-\infty$
☐	∞
☐	0

Click to Save Answer & Move to Next Question



BC190406254: TAMOOR JAMIL

Time Left

89
sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 11:12 PM, 14 December 2022

Question # 8 of 10 (Start time: 11:20:21 PM, 14 December 2022)

Total Marks: 1

The leading coefficient in the differential equation

$$3x \frac{d^2 y}{dx^2} - 7x \frac{dy}{dx} + 5y = 0$$

Select the correct option

Reload Math Equations

☐	-7
☐	3
☐	5
☐	7

Click to Save Answer & Move to Next Question



BC210420813: MUHAMMAD SAQLAIN

Time Left 88
sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 10:04 PM, 14 December 2022

Question # 8 of 10 (Start time: 10:13:44 PM, 14 December 2022)

Total Marks: 1

The differential equation

$$\frac{d^2 u}{dx^2} - 2xu + \sin x = 0$$

is

Select the correct option

Reload Math Equations

- | | |
|-----------------------|-------------------------------|
| ☐ | linear and non-homogenous |
| ☐ | linear and homogenous |
| ☐ | non-linear and non-homogenous |
| ☐ | non-linear and homogenous |

Click to Save Answer & Move to Next Question

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 10:49 PM, 14 December 2022

Question # 1 of 10 (Start time: 10:49:24 PM, 14 December 2022)

Total Marks: 1

The differential equation

$$\frac{du}{dx} + \frac{2}{x}dx = 3 - \frac{6}{x}$$

has an integrating factor

Select the correct option

Reload Math Equations

☐	x^2
☐	$-\frac{1}{x}$
☐	$\frac{1}{x}$
☐	$\frac{1}{x^2}$

Click to Save Answer & Move to Next Question



BC210420813: MUHAMMAD SAQLAIN

Time Left 88
sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 10:04 PM, 14 December 2022

Question # 9 of 10 (Start time: 10:15:43 PM, 14 December 2022)

Total Marks: 1

The two families of curves are _____ at the point of intersection.

Select the correct option

☐	orthogonal
☐	parallel

Click to Save Answer & Move to Next Question



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3



BC210403078: ABDUL RASHEED

Time Left 85 sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 10:59 PM, 14 December 2022

Question # 7 of 10 (Start time: 11:07:12 PM, 14 December 2022)

Total Marks: 1

The differential equation

$$\frac{d^3 y}{dx^3} + y^2 = x^4$$

is _____

Select the correct option

Reload Math Equations

- | | |
|-----------------------|--------------------------------|
| ☐ | Non-homogeneous and linear |
| ☐ | Homogeneous and non linear |
| ☐ | Non homogeneous and non-linear |
| ☐ | None of these |

Click to Save Answer & Move to Next Question





BC200404637: FATIMA ASHRAF

Time Left 89
sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 08:35 PM, 14 December 2022

Question # 10 of 10 (Start time: 08:47:56 PM, 14 December 2022)

Total Marks: 1

The differential equation

$$\frac{d^2 u}{dx^2} - 2xu + \sin x = 0$$

is

Select the correct option

Reload Math Equations

- | | |
|-----------------------|-------------------------------|
| ☐ | non-linear and homogenous |
| ☐ | linear and non-homogenous |
| ☐ | non-linear and non-homogenous |
| ☐ | linear and homogenous |

Click to Save Answer & Move to Next Question

A function satisfying the differential equation on the interval I whose graph passes through (x_0, y_0) such that the slope of the curve at the point is the number y'_0 is called solution of the _____ value problem.

The correct option

initial

boundary

Question # 6 of 10 (Start time: 10:05:58 PM, 14 December 2022)

The differential equation

$$y(1 + 2xy)dx + x(1 - 2xy)dy = 0$$

is

Select the correct option



Relo

☐	Separable
☐	None of these
☐	Homogeneous
☐	Exact

Activate Windows
Go to Settings to activate Windows

BC190407612: MUZAMMAL RASHEED

Time Left 78
sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 08:47 PM, 14 December 2022

Question # 5 of 10 (Start time: 08:56:04 PM, 14 December 2022)

Total Marks: 1

In exponential model for the population growth

$$P(t) = P_0 e^{kt} \quad \text{If } k > 0, \text{ then } \lim_{t \rightarrow \infty} P(t) =$$

Select the correct option

Reload Math Equations

☐	∞
☐	$-\infty$
☐	0
☐	1

Click to Save Answer & Move to Next Question





BC190406254: TAMOOR JAMIL

Time Left 54 sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 11:12 PM, 14 December 2022

Question # 10 of 10 (start time: 11:23:04 PM, 14 December 2022)

Total Marks: 1

In exponential growth for population growth

$$N(t) = N_0 e^{mt}$$

If $m < 0$ then

$$\lim_{t \rightarrow \infty} N(t) =$$

Select the correct option

Reload Math Equations

☐	$-\infty$
☐	∞
☐	1
☐	0

Click to Save Answer & Move to Next Question



BC210420813: MUHAMMAD SAQLAIN

Time Left 88
sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 10:04 PM, 14 December 2022

Question # 10 of 10 (Start time: 10:17:42 PM, 14 December 2022)

Total Marks: 1

In the logistic equation

$$\frac{1}{p} \frac{dp}{dt} = a - bp$$

the term

$$-bp^2$$

can be interpreted as

Select the correct option

Reload Math Equations

☐	unconstraint term
☐	inhibition term
☐	self-abandonment term
☐	constraint term

Click to Save Answer & Move to Next Question



BC190407612: MUZAMMAL RASHEED

Time Left 88
sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 08:47 PM, 14 December 2022

Question # 1 of 10 (Start time: 08:47:48 PM, 14 December 2022)

Total Marks: 1

If we substitute

$$u = \frac{y}{x}$$

in differential equation

$$x \frac{dy}{dx} - y = \frac{x^3}{y} e^{\frac{y}{x}}$$

then the given differential equation can be simplified to

Select the correct option

Reload Math Equations

☐	$ue^{(-u)} du = dy$
☐	$ue^u du = dx$
☐	$ue^u du = dy$
☐	$ue^{(-u)} du = dx$

Click to Save Answer & Move to Next Question





BC190406254: TAMOOR JAMIL

Time Left 89 sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 11:12 PM, 14 December 2022

Question # 1 of 10 (Start time: 11:12:08 PM, 14 December 2022)

Total Marks: 1

The differential equation

$$y(1 + 2xy)dx + x(1 - 2xy)dy = 0$$

is

Select the correct option

Reload Math Equations

☐	Exact
☐	Homogeneous
☐	None of these
☐	Separable

Click to Save Answer & Move to Next Question



BC210420813: MUHAMMAD SAQLAIN

Time Left 89
sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 10:04 PM, 14 December 2022

Question # 1 of 10 (Start time: 10:04:53 PM, 14 December 2022)

Total Marks: 1

The Wronskian of two functions

$$f(x)$$

and

$$g(x)$$

is

Select the correct option

Reload Math Equations

☐	$f(x)g'(x) - f'(x)g(x)$
☐	$f(x)g'(x) + f'(x)g(x)$
☐	$f'(x)g'(x) + f(x)g(x)$
☐	$f'(x)g'(x) - f(x)g(x)$

Click to Save Answer & Move to Next Question



Quiz

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BC190407612: MUZAMMAL RASHEED

Time Left 85 sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 08:47 PM, 14 December 2022

Question # 9 of 10 (Start time: 09:01:33 PM, 14 December 2022)

Total Marks: 1

The leading coefficient in the differential equation

$$3x \frac{d^2 y}{dx^2} - 7x \frac{dy}{dx} + 5y = 0$$

Select the correct option

Reload Math Equations

☐	3
☐	5
☐	-7
☐	7

Click to Save Answer & Move to Next Question



BC200400580: USMAN SALEEM

Time Left 89 sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 10:49 PM, 14 December 2022

Question # 5 of 10 (Start time: 10:53:26 PM, 14 December 2022)

Total Marks: 1

The leading coefficient in the differential equation

$$3x \frac{d^2 y}{dx^2} - 7x \frac{dy}{dx} + 5y = 0$$

Select the correct option

Reload Math Equations

- | | |
|-----------------------|----|
| ☐ | 7 |
| ☐ | 5 |
| ☐ | -7 |
| ☐ | 3 |

Click to Save Answer & Move to Next Question



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BC210403078: ABDUL RASHEED

Time Left 89
sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 10:59 PM, 14 December 2022

Question # 4 of 10 (Start time: 11:03:32 PM, 14 December 2022)

Total Marks: 1

The differential equation

$$\frac{du}{dx} + \frac{2}{x}u = 3 - \frac{6}{x}$$

has an integrating factor

Select the correct option

Reload Math Equations

☐	$\frac{1}{x^2}$
☐	$-\frac{1}{x}$
☐	$\frac{1}{x}$
☐	x^2

Click to Save Answer & Move to Next Question





BC190406254: TAMOOR JAMIL

Time Left

89
sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 11:12 PM, 14 December 2022

Question # 7 of 10 (Start time: 11:18:31 PM, 14 December 2022)

Total Marks: 1

Wronskian of 1 and x^2 , $W(1, x^2) = \underline{\hspace{2cm}}$

Select the correct option

Reload Math Equations

- | | |
|-----------------------|---------------|
| ☐ | x |
| ☐ | 2x |
| ☐ | 0 |
| ☐ | None of these |

Click to Save Answer & Move to Next Question



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BC210403078: ABDUL RASHEED

Time Left 87 sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 10:59 PM, 14 December 2022

Question # 5 of 10 (Start time: 11:04:32 PM, 14 December 2022)

Total Marks: 1

For 2nd order linear non-homogeneous differential equation

$$a_2(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

, which of the following is NOT the pair of boundary condition _____

Select the correct option

Reload Math Equations

☐	$y'(a) = y_0', \quad y(b) = y_1$
☐	$y'(a) = y_0', \quad y'(b) = y_0'$
☐	$y'(a) = y_0', \quad y(b) = y_1$
☐	$y'(a) = y_0', \quad y'(b) = y_1'$

Click to Save Answer & Move to Next Question



Question # 8 of 10 (Start time: 10:58:12 PM, 14 December 2022)

Total Marks: 1

In exponential model for the population growth

$$P(t) = P_0 e^{kt} \text{ If } k > 0, \text{ then } \lim_{t \rightarrow \infty} P(t) =$$

Select the correct option

Reload Math Equations

☐	∞
☐	0
☐	1
☐	$-\infty$

Click to Save Answer & Move to Next Question



BC190407612: MUZAMMAL RASHEED

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time

Question # 10 of 10 (Start time: 09:02:52 PM, 14 December 2022)

The homogeneous linear differential equations always possess the _____ solution.

Select the correct option

- | | |
|-----------------------|-------------|
| ☐ | trivial |
| ☐ | non-trivial |

Click to Save Answer



Question # 10 of 10 (Start time: 08:47:27 PM, 14 December 2022)

A modification of the nonlinear logistic differential equation

has been used in the study of

$$\frac{1}{p} \frac{dp}{dt} = a - b \ln p$$

Select the correct option

- ☐ revenue growth of commercial product
- ☐ solid tumor
- ☐ actuarial prediction
- ☐ all of these

Question # 2 of 10 (Start time: 10:00:39 PM, 14 December 2022)

The logistic equation, $\frac{dP}{dt} = P(a - bP)$, can be identified as a nonlinear equation

Select the correct option

 Reload☐ integrable☐ separableActivate Windows
Go to Settings to activate Windows.

Question # 5 of 10 (Start time: 10:05:03 PM, 14 December 2022)

The differential equation of orthogonal trajectory to the family of curves

$$x - 2y = c$$

is

Select the correct option



Relax



$$\frac{dy}{dx} = 2$$



$$\frac{dy}{dx} = -2$$



$$\frac{dy}{dx} = \frac{1}{2}$$



$$\frac{dy}{dx} = -\frac{1}{2}$$

Activate Windows

Go to Settings to activate Windows



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Question # 7 of 10 (Start time: 05:20:46 PM, 14 December 2022)

Total Marks: 1

In the logistic equation:

$$\frac{1}{p} \frac{dp}{dt} = a - bp$$

the term:

$$-bp^2$$

can be interpreted as

Select the correct option

 Reload Math Equations

☐	self-abandonment term
☐	inhibition term
☐	unconstraint term
☐	constraint term

Click to Save Answer & Move to Next Question

If Wronskian $W(h,z)(x)=0$ then

Select the correct option

 Reload Math Equations

☐

$h(x)$ and $z(x)$ are linearly independent.

☐

$h(x)$ and $z(x)$ are linearly dependent.

Click to Save Answer & Move to Next Question



BC200414970: AMNA

Time Left 69
sec(s)

MTH401 – Differential Equations (Quiz#1)

Quiz Start Time: 04:42 PM, 14 December 2022

Question # 7 of 10 (Start time: 04:47:36 PM, 14 December 2022)

Total Marks: 1

Orthogonal trajectories occur naturally in the area(s) of_____.

Select the correct option



(ii) fluid dynamics



(iii) electricity and magnetism



(iv) All (i), (ii) and (iii).



(i) physics

Click to Save Answer & Move to Next Question



Question # 7 of 10 (Start time: 10:56:43 PM, 14 December 2022) Total Marks: 1

The Wronskian of two functions

$$f(x)$$

and

$$g(x)$$

is

[Reload Math Equations](#)

Select the correct option

☐

$$f'(x)g''(x) - f(x)g'(x)$$

☐

$$f'(x)g''(x) + f(x)g'(x)$$

☐

$$f(x)g''(x) - f'(x)g'(x)$$

☐

$$f(x)g''(x) + f'(x)g'(x)$$

[Click to Save Answer & Move to Next Question](#)

BC200400580: USMAN SALEEM

Time Left 88 sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 10:49 PM, 14 December 2022

Question # 3 of 10 (Start time: 10:51:20 PM, 14 December 2022)

Total Marks: 1

The differential equation

$$y(1 + 2xy)dx + x(1 - 2xy)dy = 0$$

is

Select the correct option

Reload Math Equations

☐

Homogeneous

☐

None of these

☐

Exact

☐

Separable

Click to Save Answer & Move to Next Question

BC200400580: USMAN SALEEM

Time Left 88 sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 10:49 PM, 14 December 2022

Question # 10 of 10 (Start time: 11:00:58 PM, 14 December 2022)

Total Marks: 1

The differential equation of orthogonal trajectory to the family of curves

$$x^2 + y^2 = C^2$$

is _____

Select the correct option

Reload Math Equations

☐

$$\frac{dy}{dx} = \frac{x}{y}$$

☐

$$\frac{dy}{dx} = -\frac{x}{y}$$

☐

$$\frac{dy}{dx} = \frac{y}{x}$$

☐

$$\frac{dy}{dx} = -\frac{y}{x}$$



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BC210403078: ABDUL RASHEED

Time Left 90 sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 10:59 PM, 14 December 2022

Question # 1 of 10 (Start time: 10:59:01 PM, 14 December 2022)

Total Marks: 1

The other name of Verhulst Pearl is ...

Select the correct option

Reload Math Equations

☐	Gaussian model
☐	Logistic model
☐	Epidemic model
☐	Euler model

Click to Save Answer & Move to Next Question



MTH401 – Differential Equations (Quiz#1)

Quiz

Question # 2 of 10 (Start time: 10:08:25 PM, 14 December 2022)

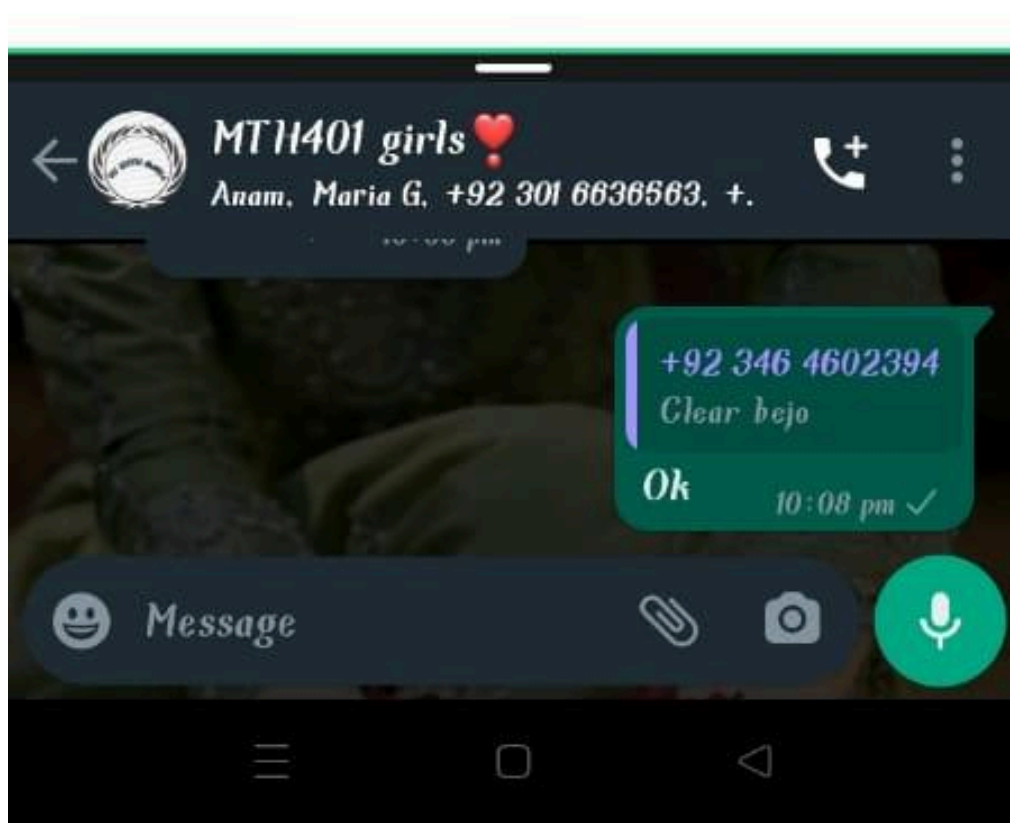
In exponential model for the population growth

$$P(t) = P_0 e^{kt} \text{ If } k < 0, \text{ then } \lim_{t \rightarrow \infty} P(t) =$$

Select the correct option

- | | |
|-----------------------|-----------|
| ☐ | 1 |
| ☐ | ∞ |
| ☐ | 0 |
| ☐ | $-\infty$ |

Click to



Question # 6 of 10 (Start time: 05:19:54 PM, 14 December 2022)

Total Marks: 1

In the population growth model, the solution of the differential equation is given by _____.

Select the correct option

 Reload Math Equations

☐ $A(t) = A_0/2$

☐ $A(t) = A_0/2$

☐ $A(t) = ke^{kt}$

☐ $A(t) = A_0e^{kt}$

Click to Save Answer & Move to Next Question



Quiz

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BC190407612: MUZAMMAL RASHEED

Time L

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 08:47 PM, 14 Dec

Question # 6 of 10 (start time: 08:57:35 PM, 14 December 2022)

T

A function satisfying the differential equation on the some interval I , containing a and b , whose graph passes through two points (a, y_0) and (b, y_1) is called solution of the _____ value problem.

select the correct option

Reload Math

☐	boundary
☐	initial

Click to Save Answer & Move to Next Q





BC210420813: MUHAMMAD SAQLAIN

Time Left 89
sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 10:04 PM, 14 December 2022

Question # 2 of 10 (Start time: 10:06:04 PM, 14 December 2022)

Total Marks: 1

The differential equation of orthogonal trajectory to the family of curves

$$x - 2y = c$$

is

Select the correct option

Reload Math Equations

☐	$\frac{dy}{dx} = \frac{1}{2}$
☐	$\frac{dy}{dx} = -2$
☐	$\frac{dy}{dx} = -\frac{1}{2}$
☐	$\frac{dy}{dx} = 2$

Click to Save Answer & Move to Next Question



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BC210403078: ABDUL RASHEED

Time Left 90
sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 10:59 PM, 14 December 2022

Question # 3 of 10 (start time: 11:01:10 PM, 14 December 2022)

Total Marks: 1

The two families of curves are _____ at the point of intersection.

Select the correct option

☐ parallel☐ orthogonal

Click to Save Answer & Move to Next Question





BC210420813: MUHAMMAD SAQLAIN

Time Left
88
sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 10:04 PM, 14 December 2022

Question # 4 of 10 (Start time: 10:07:43 PM, 14 December 2022)

Total Marks: 1

If

$$a_n(x) = 0$$

in the differential equation

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

for some

$$x \in I$$

then solution of initial value problem may NOT be_____.

Select the correct option

Reload Math Equations

☐	unique
☐	exist
☐	exist and unique
☐	none of these

Click to Save Answer & Move to Next Question



BC210420813: MUHAMMAD SAQLAIN

Time Left 87
sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 10:04 PM, 14 December 2022

Question # 7 of 10 (start time: 10:12:19 PM, 14 December 2022)

Total Marks: 1

When the arbitrary constant of the general solution of a differential equation takes some unique value, then the solution becomes theof the equation.

Select the correct option

Reload Math Equations

☐	implicit solution
☐	explicit solution
☐	none of these
☐	particular solution

Click to Save Answer & Move to Next Question



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BC210403078: ABDUL RASHEED

Time Left 90
sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 10:59 PM, 14 December 2022

Question # 8 of 10 (Start time: 11:07:54 PM, 14 December 2022)

Total Marks: 1

The general solution of non-homogenous equation is,

Select the correct option

Reload Math Equations

☐	$y_p - y_c$
☐	$y_c * y_p$
☐	$y_c - y_p$
☐	$y_p + y_c$

Click to Save Answer & Move to Next Question





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3



BC210403078: ABDUL RASHEED

Time Left 90 sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 10:59 PM, 14 December 2022

Question # 10 of 10 (Start time: 11:10:01 PM, 14 December 2022)

Total Marks: 1

Wronskian of 1 and x^2 , $W(1, x^2) = \underline{\hspace{2cm}}$

Select the correct option

Reload Math Equations

☐	None of these
☐	$2x$
☐	x
☐	0

Click to Save Answer & Move to Next Question

BC200400580: USMAN SALEEM

Time Left 88 sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 10:49 PM, 14 December 2022

Question # 4 of 10 (Start time: 10:52:13 PM, 14 December 2022)

Total Marks: 1

In exponential model for the population growth

$$P(t) = P_0 e^{kt} \text{ If } k < 0, \text{ then } \lim_{t \rightarrow \infty} P(t) =$$

Select the correct option

Reload Math Equations

- | | |
|-----------------------|-----------|
| ☐ | $-\infty$ |
| ☐ | 1 |
| ☐ | 0 |
| ☐ | ∞ |

Click to Save Answer & Move to Next Question

Question # 4 of 10 (**Start time: 10:10:21 PM, 14 December 2022**)

A set of n functions

$$\{f_1(x), f_2(x), f_3(x), \dots, f_n(x)\}$$

is _____ if at least one of them can be expressed as a linear combination of the remaining.

Select the correct option

- | | |
|-----------------------|--------------------|
| ☐ | linear dependent |
| ☐ | linear independent |

Click to Se

Question # 9 of 10 (start time: 11:04:41 PM, 14 December 2022)

Total Marks: 1

The two families of curves are _____ at the point of intersection.

Select the correct option

☐ orthogonal☐ parallel[Click to Save Answer & Move to Next Question](#)



Quiz

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Question # 6 of 10 (Start time: 10:12:33 PM, 14 December 2022)

In the logistic equation

$$\frac{1}{p} \frac{dp}{dt} = a - bp$$

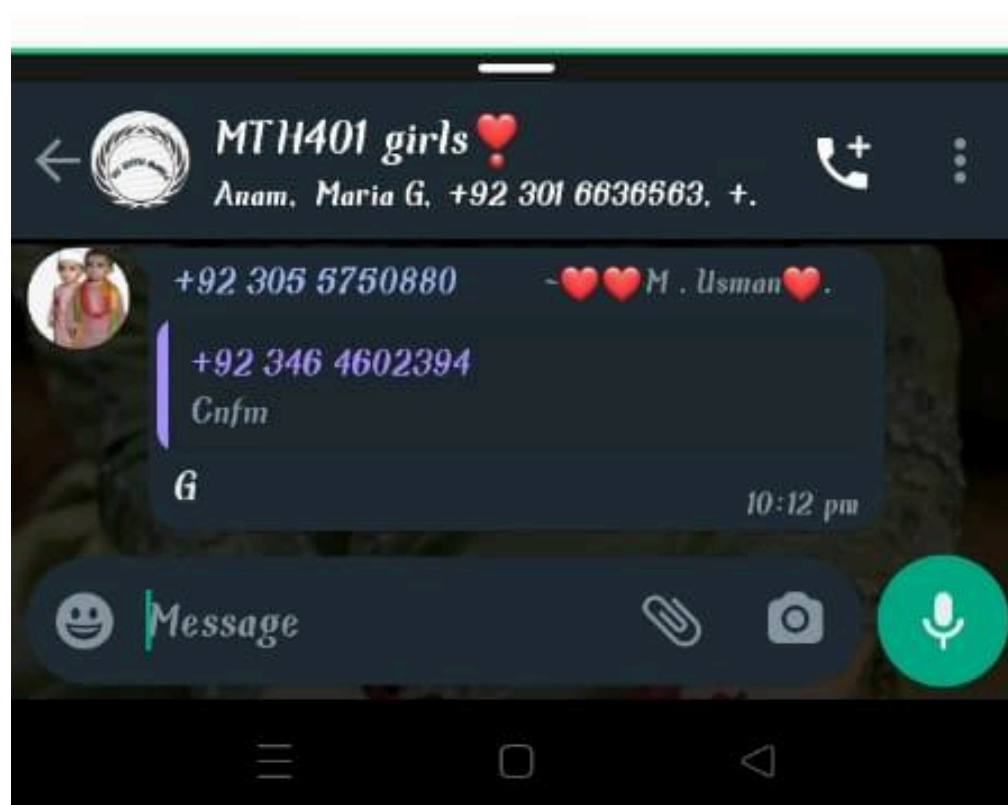
the term

$$-bp^2$$

can be interpreted as

Select the correct option

- | | |
|-----------------------|-----------------------|
| ☐ | self-abandonment term |
| ☐ | inhibition term |
| ☐ | constraint term |
| ☐ | unconstraint term |



10:13

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Quiz

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Differential Equations (Quiz#1)

Quiz Start Time: 10:07 PM, 14 Dece

of 10 (Start time: 10:13:17 PM, 14 December 2022)

To

A function satisfying the differential equation on the some interval I , containing a and b , whose graph passes through two points (a, y_0) and (b, y_1) is called solution of the _____ value problem.

Correct option

Reload Math Engine

initial

boundary

Click to Save Answer & Move to Next Q

MT11401 girls ❤️

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+92 346 4602394

-lq o. 🌹🌹

+92 305 5750880

B

B

10:13 pm

Message

10:07

×

Quiz

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RC204040761: GAMAN HUNTER

Time Left: 00:00:00

MT11401- Differential equations (Quiz#1)

Quiz start time: 10:07 PM, 14 December 2022

Question # 1 of 10 (Start time: 10:07:23 PM, 14 December 2022)

Total Marks: 10

The logistic equation, $\frac{dP}{dt} = P(a - bP)$, can be identified as a nonlinear equation that is _____.

Select the correct option

Reload Math equation

☐	separable
☐	integrable

Click to Save Answer & Move to Next Question

←

MT11401 girls ❤️

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M start kr rahe hn

10:06 pm ✓

1 Unread Message

+92 346 4602394

~lqo. 🌹🌹

Message



Quiz

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MTH401 - Differential Equations (Quiz#1)

Question # 10 of 10 (Start time: 10:15:28 PM, 14 December 2022)

The differential equation of orthogonal trajectory to the family of curves

$$x - 2y = c$$

is

Select the correct option



$$\frac{dy}{dx} = 2$$



$$\frac{dy}{dx} = -2$$



$$\frac{dy}{dx} = -\frac{1}{2}$$



$$\frac{dy}{dx} = \frac{1}{2}$$



MTH401 girls ❤️

Anam, Maria G, +92 301 6636563, +.

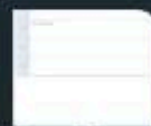


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You

Photo



1

10:15 pm



Message





0 (Start time: 05:01:33 PM, 14 December 2022)

If $A(t)$ is the amount of a radioactive substance present at time t , the the rate of change of $A(t)$ with respect to t is given by _____.

Correct option

Reload

$$\frac{A_0}{2}$$

$$dA = dt$$

$$\frac{dA}{dt} = kA; k \text{ is a constant of proportionality}$$

$$A_0 e^t$$

Click to Save Answer & Move to Next Question

Question # 5 of 10 (Start time: 07:05:55 PM, 14 December 2022)

Total Marks: 1

The other name of Verhulst Pearl is ...

Select the correct option

 Reload Math Equations

Euler model



Epidemic model



Logistic model



Gaussian model

Click to Save Answer & Move to Next Question

Question # 7 of 10 (Start time: 07:07:42 PM, 14 December 2022)

Total Marks: 1

The differential equation.

$$\frac{du}{dx} + \frac{2}{x}dx = 3 - \frac{6}{x}$$

has an integrating factor.

Select the correct option

 Reload Math Equations


$$\frac{1}{x^2}$$



$$-\frac{1}{x}$$



$$\frac{1}{x}$$



$$x^2$$

Click to Save Answer & Move to Next Question



Quiz

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BC200414970: AMNA

Time Left

85
sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 04:42 PM, 14 December 2022

Question # 4 of 10 (start time: 04:45:17 PM, 14 December 2022)

Total Marks: 1

The differential equation

$$\frac{du}{dx} + \frac{2}{x}u = 3 - \frac{6}{x}$$

has an integrating factor

Select the correct option

Reload Math Equations

☐	$\frac{1}{x^2}$
☐	x^2
☐	$-\frac{1}{x}$
☐	$\frac{1}{x}$

Click to Save Answer & Move to Next Question





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3



BC210403078: ABDUL RASHEED

Time Left 88 sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 10:59 PM, 14 December 2022

Question # 2 of 10 (start time: 10:59:45 PM, 14 December 2022)

Total Marks: 1

If X is the remaining amount of the substance A at any time t then, $\frac{dX}{dt} = kX$, where $k < 0$ because X is ____.

Select the correct option

Reload Math Equations

☐	increasing
☐	decreasing

Click to Save Answer & Move to Next Question





BC190406254: TAMOOR JAMIL

Time Left 88 sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 11:12 PM, 14 December 2022

Question # 6 of 10 (Start time: 11:17:16 PM, 14 December 2022)

Total Marks: 1

In the logistic equation

$$\frac{1}{p} \frac{dp}{dt} = a - bp$$

the term

$$-bp^2$$

can be interpreted as

Select the correct option

Reload Math Equations

☐	inhibition term
☐	unconstraint term
☐	self-abandonment term
☐	constraint term

Click to Save Answer & Move to Next Question

Question # 8 of 10 (Start time: 05:21:38 PM, 14 December 2022)

Total Marks: 1

Wronskian

$$W(1, e^x)$$

= _____

Select the correct option

 Reload Math Equations

☐	0
☐	e^x
☐	-1
☐	1

Click to Save Answer & Move to Next Question

Question # 9 of 10 (Start time: 05:23:21 PM, 14 December 2022)

Total Marks: 1

A set of functions

$$\{f_1(x), f_2(x), f_3(x), \dots, f_n(x)\}$$

is said to be _____ on an interval I if there exist constants

$$c_1, c_2, c_3, \dots, c_n$$

are all zero, such that

$$c_1 f_1(x) + c_2 f_2(x) + \dots + c_n f_n(x) = 0, \quad \forall x \in I$$

Select the correct option



linear independent

☐

linear dependent

☐

Click to Save Answer & Move to Next Question



In first order chemical reaction ,

Select the correct option

☐	the molecule of the substance makes smaller molecules.
☐	None of them
☐	the molecules of the substance remain same.
☐	the molecule of the substance makes large molecules.

C





Question # 5 of 10 (Start time: 05:00:19 PM, 14 December 2022)

The leading coefficient in the differential equation

$$3x \frac{d^2 y}{dx^2} - 7x \frac{dy}{dx} + 5y = 0$$

Select the correct option

☐	7
☐	-7
☐	3
☐	5

Click to Save A



Quiz

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BC190407612: MUZAMMAL RASHEED

Time Left 82
sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 08:47 PM, 14 December 2022

Question # 8 of 10 (start time: 08:59:56 PM, 14 December 2022)

Total Marks: 1

When the arbitrary constant of the general solution of a differential equation takes some unique value, then the solution becomes theof the equation.

Select the correct option

Reload Math Equations

☐	none of these
☐	explicit solution
☐	implicit solution
☐	particular solution

Click to Save Answer & Move to Next Question





BC210420813: MUHAMMAD SAQLAIN

Time Left 81
sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 10:04 PM, 14 December 2022

Question # 6 of 10 (Start time: 10:10:56 PM, 14 December 2022)

Total Marks: 1

A set of n functions

$$\{f_1(x), f_2(x), f_3(x), \dots, f_n(x)\}$$

is _____ if at least one of them can be expressed as a linear combination of the remaining.

Select the correct option

Reload Math Equations



linear independent



linear dependent

Click to Save Answer & Move to Next Question

If Wronskian $W(h,z)(x)=0$ then

Select the correct option

☐ $h(x)$ and $z(x)$ are linearly independent.

☐ $h(x)$ and $z(x)$ are linearly dependent.



BC190406254: TAMOOR JAMIL

Time Left 88 sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 11:12 PM, 14 December 2022

Question # 2 of 10 (Start time: 11:13:14 PM, 14 December 2022)

Total Marks: 1

Let:

$$f_1(x) = e^{m_1 x}$$

and:

$$f_2(x) = e^{m_2 x}$$

are linearly independent because

$$c_1 f_1(x) + c_2 f_2(x) = 0$$

iff

Select the correct option

Reload Math Equations

☐	$c_1 = c_2 = 0, m_1 \neq m_2$
☐	$c_1 = c_2 \neq 0, m_1 = m_2$
☐	$c_1 = c_2 \neq 0, m_1 \neq m_2$
☐	$c_1 = c_2 = 0, m_1 = m_2$

Click to Save Answer & Move to Next Question



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3



BC210403078: ABDUL RASHEED

Time Left 83 sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 10:59 PM, 14 December 2022

Question # 9 of 10 (Start time: 11:08:35 PM, 14 December 2022)

Total Marks: 1

The logistic equation, $\frac{dP}{dt} = P(a - bP)$, can be identified as a nonlinear equation that is _____.

Select the correct option

Reload Math Equations

☐	separable
☐	integrable

Click to Save Answer & Move to Next Question





BC200413286: MUHAMMAD ABU BAKAR

MTH401 - Differential Equations (Quiz#1)

Question # 2 of 10 (Start time: 08:52:28 PM, 14 December 2022)

In the population growth model, the solution of the differential equation is given by _____

Select the correct option

☐	$A(t) = A_0/2$
☐	$A(t) = A_0/2$
☐	$A(t) = A_0 e^{kt}$
☐	$A(t) = k e^{kt}$

Question # 8 of 10 (Start time: 11:03:28 PM, 14 December 2022)

Total Marks: 1

When the arbitrary constant of the general solution of a differential equation takes some unique value, then the solution becomes theof the equation.

Select the correct option

 Reload Math Equations

☐	none of these.
☐	explicit solution
☐	particular solution
☐	implicit solution

[Click to Save Answer & Move to Next Question](#)



Question # 7 of 10 (Start time: 05:02:33 PM, 14 December 2022)

The homogeneous linear differential equations always possess the _____ solution.

Select the correct option

☐	trivial
☐	non-trivial

Click to Save Answer

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Quiz

New Tab

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MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 07:49 PM, 14 December 2022

Question # 2 of 10 (Start time: 07:51:10 PM, 14 December 2022)

Total Marks: 1

The differential equation

$$\frac{d^2 u}{dx^2} - 2xu + \sin x = 0$$

is

Select the correct option

Reload Math Equations

☐ non-linear and non-homogenous

☐ linear and non-homogenous

☐ non-linear and homogenous

☐ linear and homogenous

Activate Windows

Click to Save Answer & Move to Next Question

Type here to search

19°C Clear

7:51 PM 12/14/2022

If

$$a_n(x) = 0$$

in the differential equation

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

for some

$$x \in I$$

then solution of initial value problem may NOT be.....

Select the correct option

Reload Math Equations

☐

exist

☐

exist and unique

☐

unique

☐

none of these

Click to Save Answer & Move to Next Question



if

$$f(x) = x$$

and

$$g(x) = x^3$$

then Wronskian of

$$f(x)$$

and

$$g(x)$$

is

Select the correct option

☐	$2x^3$
☐	$-x^3$
☐	$-2x^3$
☐	x^3

Click



Question # 7 of 10 (Start time: 08:59:10 PM, 14 December 2022)

The family of curve

$$x^2 + y^2 = C^2$$

represents a family of circles centered at _____.

Select the correct option

- | | |
|-----------------------|-------|
| ☐ | (0,0) |
| ☐ | (1,1) |
| ☐ | (1,0) |
| ☐ | (0,1) |

Click

The Wronskian of two functions

$$f(x)$$

and

$$g(x)$$

is

Select the correct option



Reload Ma

- | | |
|-----------------------|-------------------------|
| ☐ | $f'(x)g'(x) + f(x)g(x)$ |
| ☐ | $f(x)g'(x) + f'(x)g(x)$ |
| ☐ | $f(x)g'(x) - f'(x)g(x)$ |
| ☐ | $f'(x)g'(x) - f(x)g(x)$ |



12°C



ENG

Question # 3 of 10 (Start time: 10:02:12 PM, 14 Decer

If Wronskian $W(h,z)(x)=0$ then

Select the correct option

☐	$h(x)$ and $z(x)$ are linearly independent.
☐	$h(x)$ and $z(x)$ are linearly dependent.



12°C





Quiz

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MTH401 – Differential Equations (Quiz#1)

Question # 5 of 10 (Start time: 10:11:41 PM, 14 December 2022)

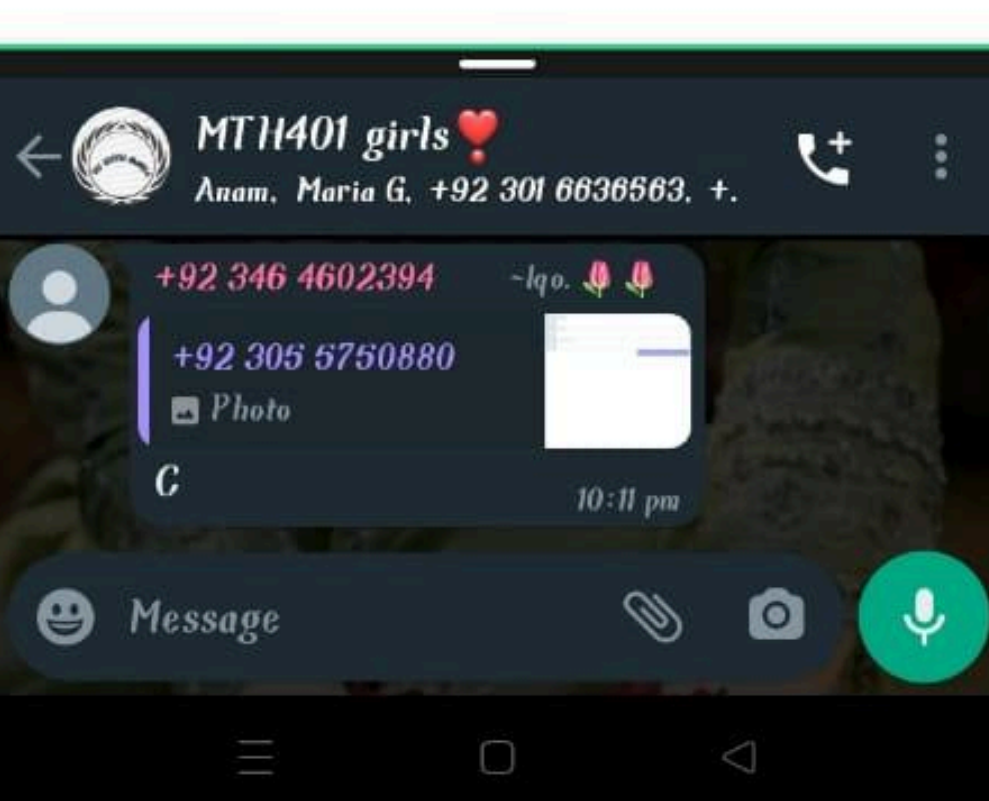
The differential equation

$$\frac{d^2u}{dx^2} - 2xu + \sin x = 0$$

is

Select the correct option

- | | |
|-----------------------|-------------------------------|
| ☐ | non-linear and homogenous |
| ☐ | non-linear and non-homogenous |
| ☐ | linear and non-homogenous |
| ☐ | linear and homogenous |



Question # 6 of 10 (Start time: 07:06:28 PM, 14 December 2022)

Total Marks: 1

The differential equation:

$$\frac{d^3 y}{dx^3} + y^2 = x^4$$

is _____.

Select the correct option

 Reload Math Equations☐

Homogeneous and non linear

☐

Non homogeneous and non-linear

☐

None of these

☐

Non-homogeneous and linear

Click to Save Answer & Move to Next Question



Quiz

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BC200414970: AMNA

Time Left 88 sec(s)

MTH401 - Differential Equations (Quiz#1)

Quiz Start Time: 04:42 PM, 14 December 2022

Question # 9 of 10 (start time: 04:50:01 PM, 14 December 2022)

Total Marks: 1

Wronskian

$$W(1, e^x)$$

= _____

Select the correct option

Reload Math Equations

☐	1
☐	0
☐	-1
☐	e^x

Click to Save Answer & Move to Next Question





MTH401 - Differential Equations (Quiz#1)

Quiz

Question # 5 of 10 (Start time: 08:55:33 PM, 14 December 2022)

The leading coefficient in the differential equation

$$3x \frac{d^2 y}{dx^2} - 7x \frac{dy}{dx} + 5y = 0$$

Select the correct option

- | | |
|-----------------------|----|
| ☐ | 5 |
| ☐ | 7 |
| ☐ | -7 |
| ☐ | 3 |

Click to



(Start time: 09:03:58 PM, 14 December 2022)

If $A(t)$ is the amount of a radioactive substance present at time t , the the rate of change of $A(t)$ with respect to t is given by _____.

option

Reload Mo

$$dA = dt$$

$$\frac{A_0}{2}$$

$$\frac{dA}{dt} = kA; k \text{ is a constant of proportionality}$$

$$A_0 e^t$$

Click to Save Answer & Move to Ne

Let

$f_1(x) = e^{m_1 x}$

and

$f_2(x) = e^{m_2 x}$

are linearly independent because

$c_1 f_1(x) + c_2 f_2(x) = 0$

iff

Select the correct option

 Reload Math Equations

- | | |
|-----------------------|----------------------------------|
| ☐ | $c_1 = c_2 \neq 0, m_1 \neq m_2$ |
| ☐ | $c_1 = c_2 = 0, m_1 \neq m_2$ |
| ☐ | $c_1 = c_2 \neq 0, m_1 = m_2$ |
| ☐ | $c_1 = c_2 = 0, m_1 = m_2$ |

Question # 4 of 10 (Start time: 07:04:33 PM, 14 December 2022)

Total Marks: 1

The homogeneous linear differential equations always possess the _____ solution.

Select the correct option

trivial

☐

non-trivial

☐[Click to Save Answer & Move to Next Question](#)



C200413286: MUHAMMAD ABU BAKAR

TH401 - Differential Equations (Quiz#1)

Question # 3 of 10 (Start time: 08:53:37 PM, 14 December 2022)

The differential equation of orthogonal trajectory to the family of curves

$$x - 2y = c$$

is

Select the correct option

☐	$\frac{dy}{dx} = \frac{1}{2}$
☐	$\frac{dy}{dx} = -2$
☐	$\frac{dy}{dx} = -\frac{1}{2}$
☐	$\frac{dy}{dx} = 2$

For 2nd order linear non-homogeneous differential equation

$$a_2(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

, which of the following is NOT the pair of boundary condition _____

Select the correct option

 Reload Math Equations

☐

$$y'(a) = y'_0, \quad y'(b) = y'_0$$

☐

$$y'(a) = y'_0, \quad y'(b) = y'_1$$

☐

$$y'(a) = y'_0, \quad y(b) = y_1$$

☐

$$y'(a) = y'_0, \quad y(b) = y_1$$

[Click to Save Answer & Move to Next Question](#)

Let

$f_1(x) = e^{m_1x}$

and

$f_2(x) = e^{m_2x}$

are linearly independent because

$c_1 f_1(x) + c_2 f_2(x) = 0$

iff

Select the correct option

 Reload Math Equations

☐	$c_1 = c_2 \neq 0, m_1 \neq m_2$
☐	$c_1 = c_2 = 0, m_1 \neq m_2$
☐	$c_1 = c_2 \neq 0, m_1 = m_2$
☐	$c_1 = c_2 = 0, m_1 = m_2$



C200413286: MUHAMMAD ABU BAKAR

MTH401 – Differential Equations (Quiz#1)

Question # 4 of 10 (Start time: 08:54:36 PM, 14 December 2022)

The differential equation

$$\frac{du}{dx} + \frac{2}{x}u = 3 - \frac{6}{x}$$

has an integrating factor

Select the correct option

☐	$\frac{1}{x}$
☐	x^2
☐	$\frac{1}{x^2}$
☐	$-\frac{1}{x}$



BC200413286: MUHAMMAD ABU BAKAR

MTH401 – Differential Equations (Quiz#1)

Question # 6 of 10 (Start time: 08:57:00 PM, 14 December 2022)

Wronskian

 $W(x, 2x)$

= _____

Select the correct option

☐	1
☐	0
☐	None of these
☐	-1



BC210425029: ABEEDA FATIMA

MTH401 - Differential Equations (Quiz#1)

Question # 10 of 10 (Start time: 08:55:58 PM, 14 December 2022)

The general solution of non-homogenous equation is,

Select the correct option



$$y_c * y_p$$



$$y_c - y_p$$



$$y_p - y_c$$



$$y_p + y_c$$



A function satisfying the differential equation on the some interval I , containing a and b , whose graph passes through two points (a, y_0) and (b, y_1) is called solution of the _____ value problem.

Select the correct option



Reload Math Equations



boundary



initial

Wronskian

$$W(1, e^x)$$

=

Select the correct option

 Reload Math Equations



1



0



$$e^x$$



-1

Click to Save Answer & Move to Next Question